ADULT TRAUMATIC CARDIAC ARREST ALGORITHM

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EUROPEAN RESUSCITATION COUNCIL©

Traumatic cardiac arrest / peri-arrest

Non-traumatic arrest likely?

YES → **ALS**

NO

START CPR

Address reversible causes simultaneously
Prioritise treatment of reversible causes over chest compressions!

- Control external catastrophic haemorrhage
- Secure airway and maximise oxygenation
- Bilateral chest decompression (thoracostomies)
- **Relieve tamponade (penetrating chest injury)**
- **Proximal vascular control (manual aortic compression)**
- Pelvic splint
- Blood products / massive haemorrhage protocol

Hypovolaemia
Hypoxia
Tension pneumothorax
Tamponade

Expertise ?
Equipment ?
Environment ?
Elapsed time ?
< 15 min

Resuscitative thoracotomy

Perform **interventions based on clinical priority,** not sequential

ROSC ?

YES

Pre-hospital:
Immediate transport to appropriate hospital

In-hospital:
Damage control surgery / resuscitation

NO

Consider termination of resuscitation

This algorithm should be used to guide the management of traumatic cardiac arrest (TCA) and peri-arrest trauma patients with impending circulatory collapse. In contrast to the sequential ALS algorithms, it is a framework for context-sensitive decisions on clinically prioritised interventions. The aim is to treat the most urgent, reversible cause based on individualised treatment (e.g. patients with a cardiac tamponade need urgent thoracotomy; patients with pelvic fractures in haemorrhagic shock but with a stable airway, may benefit more from immediate application of a pelvic binder and rapid transfer without airway intervention). ROSC – return of spontaneous circulation; ALS – advanced life support